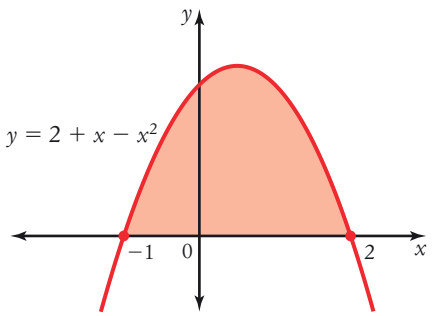
WORKSHEET

Chapter 6: Calculating physical areas

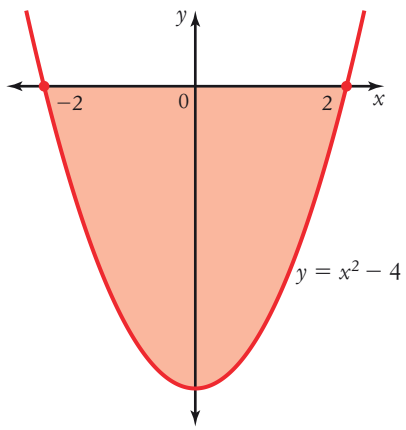
A CAS calculator is *not* to be used unless indicated.

1 Find the shaded area between the curve and the x-axis in each of the following.

a

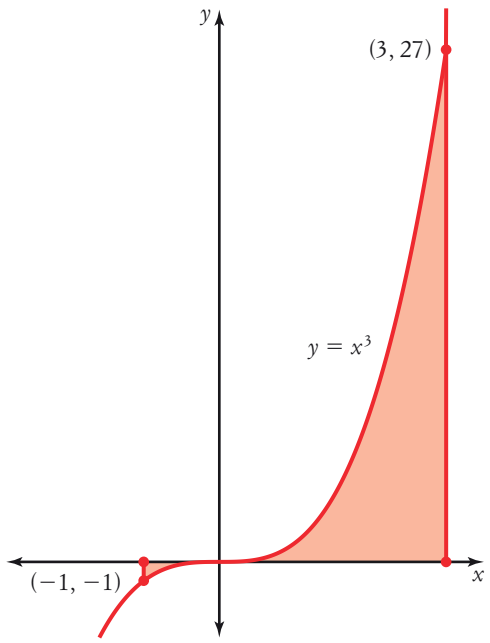


b

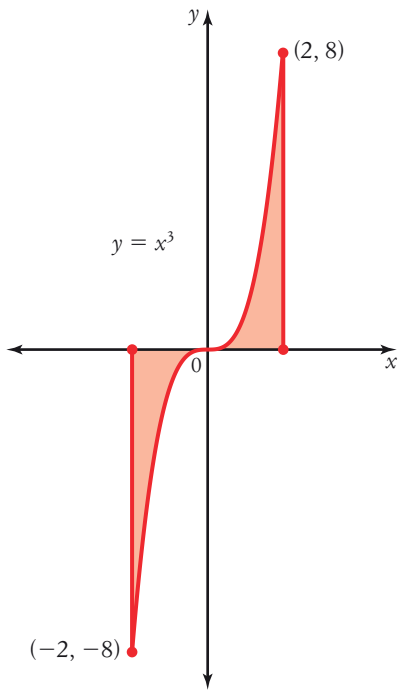


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c

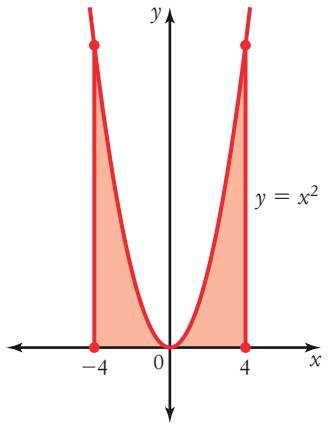


d

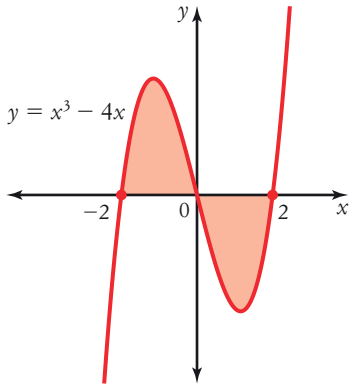


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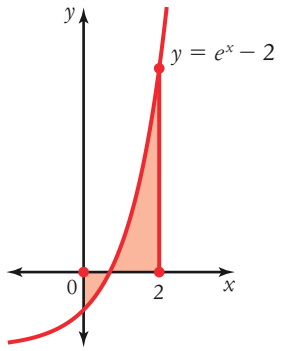
e



f

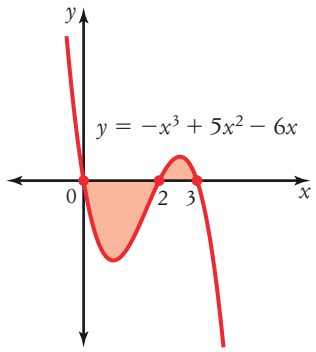


g



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h



2 By first drawing a sketch, find the area enclosed by the curve and the x-axis in each of the following cases.

a $y = 1 - x^2$

b $y = x^2 + 5x + 4$

c $y = x^2 - 2x - 3$

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d $y = -x^2 + 9x - 20$

e $y = 5x^2 - x^3 - 2x - 8$

f $y = x^3 + 3x^2 - 10x - 3$

3 CAS Use a CAS calculator to check the results of parts **b**, **d** and **f** in question **2**.

4 Find the area between the x-axis and each of the following curves for the domain given.

a $y = x^3$ between $x = 0$ and $x = 2$

b $y = x^2 + 1$ between $x = -2$ and $x = 2$

c $y = \dfrac{1}{x^2}$ between $x = 1$ and $x = 3$

d $y = \dfrac{2}{(x-3)^2}$ between $x = 0$ and $x = 1$

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e $y = \sqrt{x}$ between $x = 0$ and $x = 4$

5 CAS Use a CAS calculator to check the results of parts **a**, **c** and **e** in question **4**.

6 Find the physical area between:

a $y = (x + 2)(3 - x)(x - 1)$ and the x-axis

b $y = (x + 1)^2(x + 2)(x - 3)$ and the x-axis

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7 CAS Use a CAS calculator to find the physical area between:

a $y = e^x \sin x$ and the x-axis

b $y = \dfrac{1}{\sqrt{x-2}}$ and the x-axis

Answers

1 **a** $4\frac{1}{2}$ units^2

b $10\frac{2}{3}$ units^2

c $20\frac{1}{2}$ units^2

d 8 units^2

e $42\frac{2}{3}$ units^2

f 8 units^2

g 3.162 units^2

h $3\frac{1}{12}$ units^2

2 **a** $1\frac{1}{3}$ units^2

b $3\frac{1}{2}$ units^2

c $10\frac{2}{3}$ units^2

d $\frac{1}{6}$ units^2

e 21.08 units^2

f 94.03 units^2

4 **a** 4 units^2

b $9\frac{1}{3}$ units^2

c $\frac{2}{3}$ units^2

d $\frac{1}{3}$ units^2

e $5\frac{1}{3}$ units^2

6 **a** $21\frac{1}{12}$ units^2

b $72\frac{11}{12}$ units^2

7 **a** About 2.905 units^2

b 2 units^2

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